

MAHNOOR PARACHA

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PROFESSIONAL SUMMARY

Computer Science graduate focused on applied machine learning and full-stack development. Built a deep-learning deepfake-detection system reaching 92% accuracy and deployed a real-time GPS bus-tracking web app, working across the full workflow, data preprocessing, model training in PyTorch, frontend/backend integration, and deployment. Additional experience in NLP data pipelines and Python tutoring, with a track record of delivering client-ready work under deadline. Seeking roles in AI/ML, software development, or data.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, JavaScript

ML / Data: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, CNNs & Transfer Learning, NLP (TextBlob), Exploratory Data Analysis, Data Visualization

Web & Backend: HTML, CSS, Python (Flask), React (Vite), Tailwind CSS, Firebase (Auth, Firestore, Realtime DB), REST APIs

Tools & Platforms: Git & GitHub, Google Colab, Vercel, VS Code, Google Maps API, MS Excel / Word / PowerPoint

Design: Adobe Photoshop, Canva; posters, social media & promotional content, branding

Soft Skills: Communication, Teaching & Explaining Concepts, Teamwork, Leadership, Problem-Solving, Time Management, Attention to Detail

PROFESSIONAL EXPERIENCE

Freelance Python Developer & Tutor — Self-Employed

Sep 2024 – Present

- Taught Python, SDLC, and programming fundamentals to 5 students, improving their assessment performance and subject confidence, by building structured lesson plans and hands-on coding exercises.
- Delivered custom Python scripts and small web tools for clients, by translating requirements into working, tested code.

PROJECTS

GreenRoute — Real-Time Bus Tracking Web App

React (Vite) · Tailwind CSS · shadcn/ui · Firebase (Auth, Firestore, Realtime Database) · Google Maps API · REST APIs · Git & GitHub · Vercel

- Delivered real-time bus tracking across Sargodha routes, refreshing live bus positions every 30 seconds, by integrating GPS data with Firebase Realtime Database and the Google Maps API (Advanced Markers).
- Built a React (Vite) + Tailwind frontend giving passengers accurate routes and estimated arrival times, backed by Firebase authentication and a centralized database.
- Deployed the production app to Vercel with Git-based version control.

Deepfake Detection System — Deep Learning

Python · PyTorch · TensorFlow · OpenCV · Scikit-learn · CNNs · Transfer Learning · XceptionNet / ResNet-50 / EfficientNet · Flask · Google Colab

- Classified real vs. manipulated video at 92% accuracy on held-out test data, by fine-tuning pretrained CNNs (XceptionNet, ResNet-50, EfficientNet) in PyTorch with transfer learning.
- Built an end-to-end preprocessing pipeline — frame extraction, face cropping, and data preparation — using OpenCV and FFmpeg on the DFDC, FaceForensics++, and Celeb-DF datasets.
- Served live predictions on unseen video clips, by deploying the model behind a Flask REST API.

WhatsApp Chat Analyzer — NLP Data Pipeline & Web App

Python · Pandas · Regular Expressions · TextBlob (NLP) · WordCloud · Streamlit

- Built an end-to-end text-processing pipeline that parses raw exported chat logs with regex into a structured Pandas DataFrame, then cleans and analyzes them, message frequency, per-hour activity, and word-frequency visualization.
- Applied TextBlob NLP to compute per-message sentiment polarity and an aggregate chat-level score, with error-handled parsing to keep the pipeline robust to malformed rows.
- Shipped the analysis as an interactive Streamlit web app with file upload, live charts, and word-cloud rendering in the browser.

JARVIS — Local Voice AI Assistant (In Progress)

Python · Groq API (Llama 3.1) · Whisper (STT) · ChromaDB · SQLite · pytsx3 · FastAPI

- Building a voice-controlled desktop assistant that records mic audio, transcribes it with Whisper, and routes commands to an LLM reasoning engine (Groq Llama 3.1) with session memory of the last 20 messages.
- Implemented modular skills, SQLite-backed reminders with natural-language time parsing, a background scheduler polling every 30 seconds, live weather via the Open-Meteo API, and a timetable scraper (BeautifulSoup).

- Engineered around real constraints, swapped PyAudio for sounddevice and resolved Windows audio/ffmpeg pipeline issues to get STT working reliably on CPU.

Downloads Folder Organizer — Python Automation

Python · os · shutil

- Wrote a file-automation script that scans a directory, reads each file's extension, and sorts files into category folders (Images, Documents, Videos, Music), with an Others fallback so nothing is missed.
- Used dictionary-driven category mapping and defensive checks (`os.path.isfile`, `os.path.exists`) for safe, cross-platform, idempotent runs.

EDUCATION

University of Sargodha — B.S. Computer Science

Jun 2026

CGPA: 3.62 / 4.0 · Major: Computer Science

- *Coursework:* Data Structures & Algorithms, OOP, Artificial Intelligence, Database Systems, Computer Networks, Operating Systems, Design & Analysis of Algorithms, Compiler Construction, Digital Image Processing, Internet Architecture & Protocols, Information Systems.

CERTIFICATIONS & COURSES

- Data Analytics Consultant Simulation — Deloitte Australia (Forage Virtual Experience, Completed)
- Diploma in Data Analytics — UniAthena (Completed)
- Foundations of Data Science — Google (In Progress)
- Programming for Everybody (Python) — University of Michigan (In Progress)
- Start Writing Prompts Like a Pro — Google (Completed)
- Introduction to Networking — NVIDIA (Completed)
- How to Make Invitations Using Canva — Coursera (Completed)
- Foundations of User Experience (UX) Design — Google (In Progress)

LEADERSHIP & ACHIEVEMENTS

- Graphic Designer, Computing Media Society (CMS), University of Sargodha (2025 -26).
- Executive Head, Event Management Society (EMS), University of Sargodha (2024 -26).
- Winner, English Debate Competitions (College & School level).
- Recipient, Half Scholarship for College-level academic performance.
- Active participant in university events, workshops, and seminars (2024-26).